# Practice 6 — Combinations and Binomial Coefficients

Solutions to Selected Problems

Problems 8, 9, 11

Discrete Mathematics — QYSU

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# 1 Problem 8 — Lattice Paths Through a Given Point

**Theory Recap: Lattice Paths and Combinations**

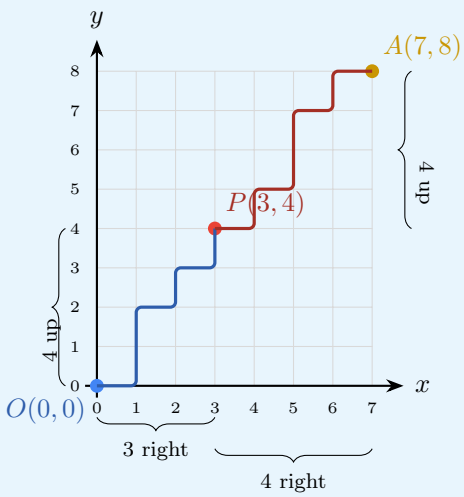
A **lattice path** from $(0,0)$ to $(m,n)$ consists of exactly $m$ steps right (R) and $n$ steps up (U), in some order. The total number of steps is $m+n$, and we must choose which $m$ of them are R-steps (the rest are U-steps):

$$\text{Number of paths from } (0,0) \text{ to } (m,n) = \binom{m+n}{m} = \binom{m+n}{n}.$$

**Paths through an intermediate point.** If a path must pass through a point $P(p,q)$ on the way from $O(0,0)$ to $A(m,n)$ (with $0 \le p \le m$ and $0 \le q \le n$), we decompose the journey into two independent segments:

$$O \to P: \binom{p+q}{p} \text{ paths}, \qquad P \to A: \binom{(m-p)+(n-q)}{m-p} \text{ paths}.$$

By the **multiplication principle**, the total count is the product.




**Problem Statement**

Find the number of lattice paths from $O(0,0)$ to $A(7,8)$ that pass through the point $P(3,4)$.

## Solution

We split the path into two independent segments at $P(3,4)$.

**Segment 1: $O(0,0) \to P(3,4)$.** We need 3 steps right and 4 steps up, for a total of 7 steps. Choose which 3 are right-steps:

$$\binom{3+4}{3} = \binom{7}{3} = \frac{7!}{3! \cdot 4!} = \frac{7 \cdot 6 \cdot 5}{3 \cdot 2 \cdot 1} = 35.$$

**Segment 2: $P(3,4) \to A(7,8)$.** From $(3,4)$ to $(7,8)$ we need $7-3=4$ steps right and

$8 - 4 = 4$ steps up, for a total of 8 steps. Choose which 4 are right-steps:

$$\binom{4+4}{4} = \binom{8}{4} = \frac{8!}{4! \cdot 4!} = \frac{8 \cdot 7 \cdot 6 \cdot 5}{4 \cdot 3 \cdot 2 \cdot 1} = 70.$$

**Multiplication principle.** The two segments are independent (any path for the first segment can be combined with any path for the second), so:

$$\text{Total paths} = 35 \times 70 = 2450.$$

**Why multiplication.** Every valid full path corresponds to exactly one first segment and one second segment. Since the choices are independent, the multiplication principle applies.

The number of lattice paths from $O(0,0)$ to $A(7,8)$ passing through $(3,4)$ is $\binom{7}{3} \cdot \binom{8}{4} = 35 \cdot 70 = \boxed{2450}$.

## 2 Problem 9 — Partitioning 6 into Three Summands

**Theory Recap: Stars and Bars / Integer Compositions and Partitions**

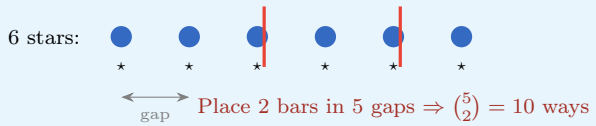
**Stars and bars** (combinations with repetition). The number of solutions to

$$x_1 + x_2 + \cdots + x_k = n, \qquad x_i \geq 0,$$

in non-negative integers is $\binom{n+k-1}{k-1}$.

If we require $x_i \geq 1$ (positive integers), substitute $y_i = x_i - 1 \geq 0$ to get $y_1 + \cdots + y_k = n - k$, giving $\binom{n-1}{k-1}$ solutions.

**Example:** $x_1 + x_2 + x_3 = 6$, $x_i \geq 1$




**Integer partitions** (unordered). A partition of $n$ into exactly $k$ parts is a way to write $n = \lambda_1 + \lambda_2 + \cdots + \lambda_k$ with $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_k \geq 1$. Order does not matter, so $4 + 1 + 1$ and $1 + 4 + 1$ are the *same* partition.

**Problem Statement**

Find the number of ways to partition the number 6 into three summands.

### Solution

**Ambiguity warning.** "Partition into three summands" can mean ordered (compositions) or unordered (partitions). We solve both interpretations below.

This problem admits two natural interpretations depending on whether the order of summands matters.

#### Interpretation 1: Ordered summands (compositions)

We seek the number of solutions to

$$x_1 + x_2 + x_3 = 6, \qquad x_i \geq 1.$$

By the stars-and-bars formula with the substitution $y_i = x_i - 1$:

$$\binom{6-1}{3-1} = \binom{5}{2} = \frac{5!}{2! \cdot 3!} = 10.$$

| # | $x_1$ | $x_2$ | $x_3$ |
|---|---|---|---|
| 1 | 4 | 1 | 1 |
| 2 | 1 | 4 | 1 |
| 3 | 1 | 1 | 4 |
| 4 | 3 | 2 | 1 |
| 5 | 3 | 1 | 2 |
| 6 | 2 | 3 | 1 |
| 7 | 1 | 3 | 2 |
| 8 | 2 | 1 | 3 |
| 9 | 1 | 2 | 3 |
| 10 | 2 | 2 | 2 |

**Interpretation 2: Unordered summands (integer partitions)**

We seek the number of ways to write $6 = \lambda_1 + \lambda_2 + \lambda_3$ with $\lambda_1 \geq \lambda_2 \geq \lambda_3 \geq 1$.

Enumerating all possibilities:

| # | **Partition** |
|---|---|
| 1 | $4+1+1$ |
| 2 | $3+2+1$ |
| 3 | $2+2+2$ |

There are exactly **3** such partitions. We verify completeness: $\lambda_1 \geq 2$ always (since $\lambda_1 \geq \lceil 6/3 \rceil = 2$). If $\lambda_1 = 4$: $\lambda_2 + \lambda_3 = 2$, so $\lambda_2 = \lambda_3 = 1$. If $\lambda_1 = 3$: $\lambda_2 + \lambda_3 = 3$ with $3 \geq \lambda_2 \geq \lambda_3 \geq 1$, giving $\lambda_2 = 2, \lambda_3 = 1$. If $\lambda_1 = 2$: $\lambda_2 + \lambda_3 = 4$ with $2 \geq \lambda_2 \geq \lambda_3 \geq 1$, giving $\lambda_2 = 2, \lambda_3 = 2$.

**Ordered** (compositions with positive parts): $\binom{5}{2} = 10$ ways.

**Unordered** (integer partitions into 3 parts): 3 ways: 4+1+1, 3+2+1, 2+2+2.

# 3 Problem 11 — Rearrangements of "matematika"

**Theory Recap: Multinomial Coefficient / Permutations with Repetition**

If we have $n$ objects where there are $k$ distinct types with multiplicities $n_1, n_2, \ldots, n_k$ (so $n_1 + n_2 + \cdots + n_k = n$), the number of distinct arrangements is the **multinomial coefficient**:

$$\binom{n}{n_1,\ n_2,\ \ldots,\ n_k} = \frac{n!}{n_1! \cdot n_2! \cdots n_k!}.$$

**Intuition.** Start with $n!$ arrangements of all objects treated as distinguishable. Then divide out the $n_i!$ internal permutations for each group of identical objects.

$$\boxed{n!\text{ total arrangements}} \quad \div \quad \boxed{n_1!} \quad \boxed{n_2!} \quad \cdots \quad \boxed{n_k!} \quad = \quad \boxed{\text{distinct arrangements}}$$

**Problem Statement**

How many different words can be formed by rearranging all the letters of the word **"matematika"**?

## Solution

**First step.** Count the letters yourself before reading the table—this is a useful habit that catches transcription errors.

**Step 1: Identify the letters and their frequencies.**

The word "matematika" has the letters:

$$\text{m} - \text{a} - \text{t} - \text{e} - \text{m} - \text{a} - \text{t} - \text{i} - \text{k} - \text{a}$$

This gives $n = 10$ letters in total. Counting each:

| Letters: | **m** | **a** | **t** | **e** | **m** | **a** | **t** | **i** | **k** | **a** |
|---|---|---|---|---|---|---|---|---|---|---|
| Position: | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |

| **Letter** | a | m | t | e | i | k |
|---|---|---|---|---|---|---|
| **Count** $n_i$ | 3 | 2 | 2 | 1 | 1 | 1 |

**Verification:** $3 + 2 + 2 + 1 + 1 + 1 = 10$ ✓

**Step 2: Apply the multinomial coefficient.**

$$\frac{10!}{3! \cdot 2! \cdot 2! \cdot 1! \cdot 1! \cdot 1!} = \frac{10!}{3! \cdot 2! \cdot 2!}$$

Computing:

$$\begin{aligned} 10! &= 3\,628\,800, \\ 3! \cdot 2! \cdot 2! &= 6 \cdot 2 \cdot 2 = 24, \\ \frac{3\,628\,800}{24} &= 151\,200. \end{aligned}$$

The number of distinct rearrangements of "matematika" is $\dfrac{10!}{3! \cdot 2! \cdot 2!} = \boxed{151\,200}$.